



CHAPTER

4

Engineering Thermodynamics I

Bahir Dar Institute of Technology,
Bahir Dar, Ethiopia

First Law of Thermodynamics



Objective of the chapter

- ❖ The objectives of this chapter is to develop and illustrate the use of the control mass and volume forms of the conservation of mass and conservation of energy principles

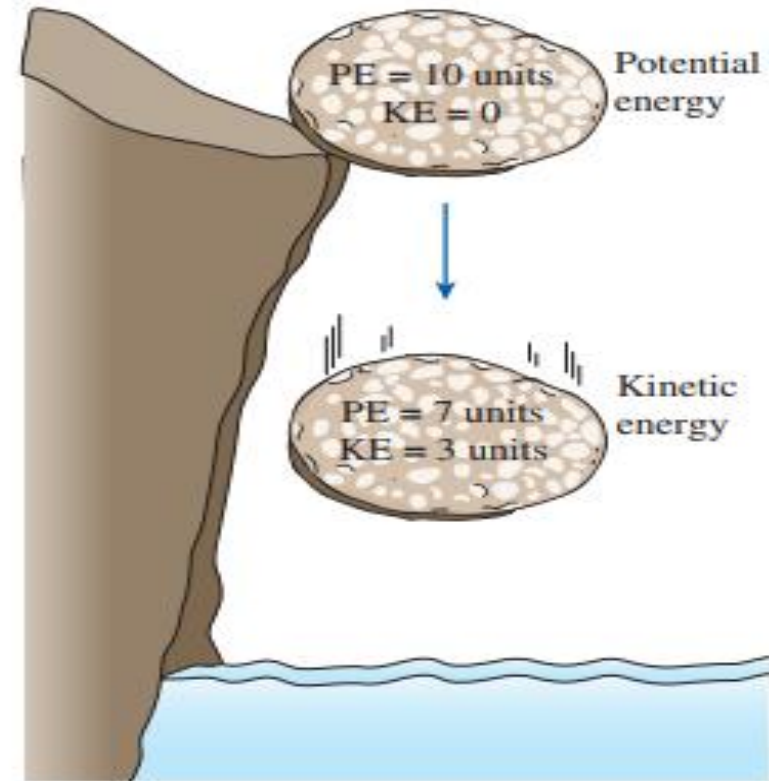


Learning outcomes

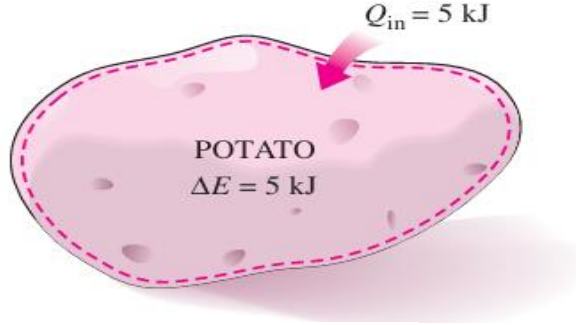
- ❖ When you complete your study of this chapter, you will be able to...
- ❖ demonstrate understanding of key concepts related to Energy and first law of Thermodynamics
- ❖ apply mass and energy balances to control mass and volumes.
- ❖ develop appropriate engineering models for control volumes, with particular attention to analyzing components commonly encountered in engineering practice such as nozzles, diffusers, turbines, compressors, heat exchangers, throttling devices, and integrated systems that incorporate two or more components.
- ❖ use property data in closed system and control volume analysis appropriately.

First Law of Thermodynamics

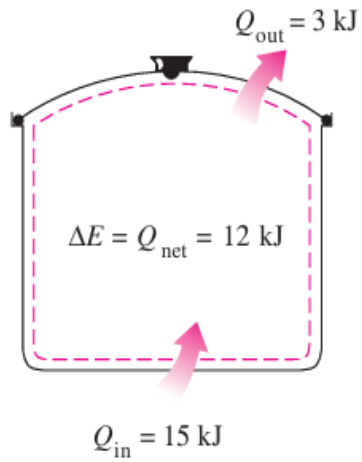
- ❖ The First Law is usually referred to as the Law of Conservation of Energy, i.e. *energy can neither be created nor destroyed, but rather transformed from one state to another.*
- ❖ The energy balance is maintained within the system being studied/defined boundary.
- ❖ The various energies associated are then being observed as they cross the boundaries of the system.



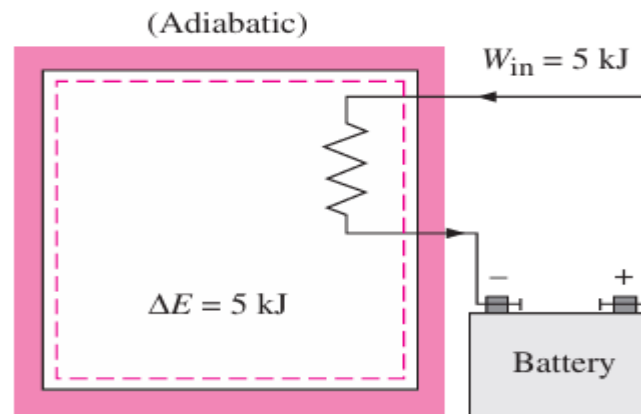
Example



The increase in the energy of a potato in an oven is equal to the amount of heat transferred to it.

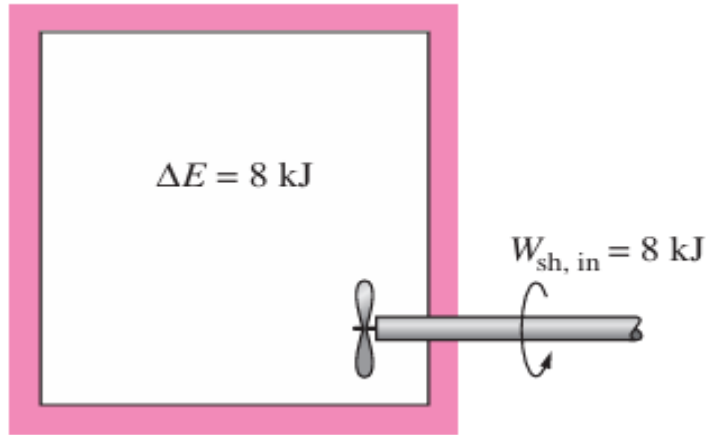


In the absence of any work interactions, the energy change of a system is equal to the net heat transfer.

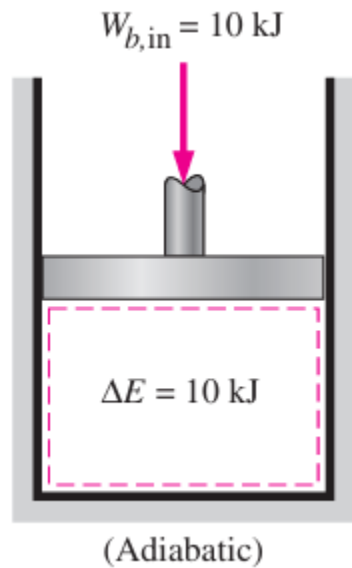


The work (electrical) done on an adiabatic system is equal to the increase in the energy of the system

(Adiabatic)



The work (shaft) done on an adiabatic system is equal to the increase in the energy of the system.



The work (boundary) done on an adiabatic system is equal to the increase in the energy of the system

Energy Balance

- The first law of thermodynamics leads to **energy balance** equation
- The conservation of energy principle can be expressed as follows:
 - ✓ The net change (increase or decrease) in the total energy of the system during a process is equal to the difference between the total energy entering and the total energy leaving the system during that process. That

is,

$$\left(\begin{array}{c} \text{Total energy} \\ \text{entering the system} \end{array} \right) - \left(\begin{array}{c} \text{Total energy} \\ \text{leaving the system} \end{array} \right) = \left(\begin{array}{c} \text{Change in the total} \\ \text{energy of the system} \end{array} \right)$$

or

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

- This relation is often referred to as the **energy balance** and is applicable to any kind of system undergoing any kind of process.

Energy Change of a System, ΔE_{system}

- The determination of the energy change of a system during a process involves the evaluation of the energy of the system **at the beginning** and at the **end of the process**, and taking their difference. That is,

$$\text{Energy change} = \text{Energy at final state} - \text{Energy at initial state}$$

or

$$\Delta E_{\text{system}} = E_{\text{final}} - E_{\text{initial}} = E_2 - E_1$$

- energy can exist in numerous forms such as internal (sensible, latent, chemical, and nuclear), kinetic, potential, electric, and magnetic, and their sum constitutes the total energy E of a system.
- In the absence of electric, magnetic, and surface tension effects (i.e., for simple compressible systems), the change in the total energy of a system during a process is the sum of the changes in its internal, kinetic, and potential energies and can be expressed as

$$\Delta E = \Delta U + \Delta KE + \Delta PE$$

where

$$\Delta U = m(u_2 - u_1)$$

$$\Delta KE = \frac{1}{2}m(V_2^2 - V_1^2)$$

$$\Delta PE = mg(z_2 - z_1)$$

Stationary Systems

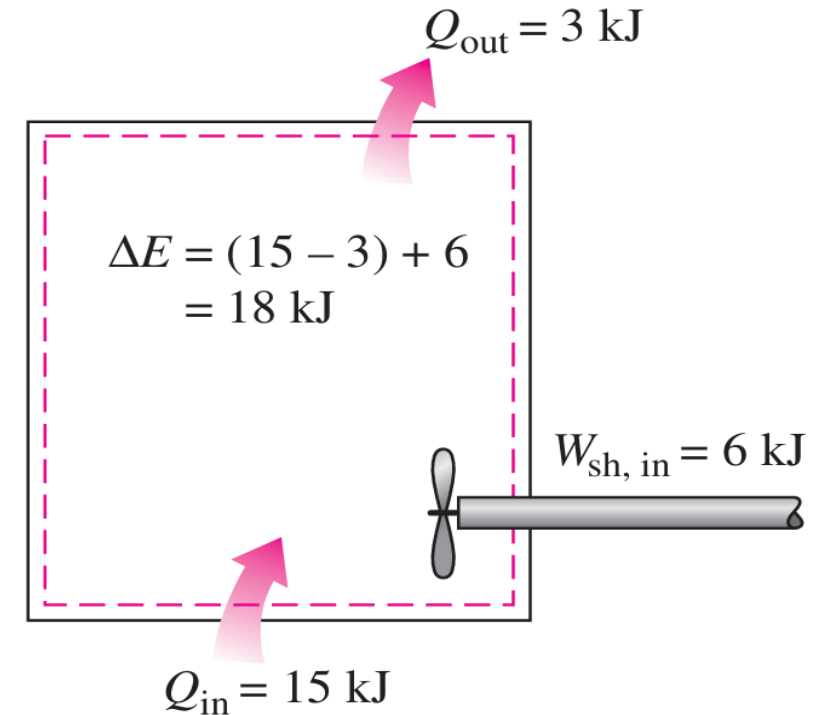
$$z_1 = z_2 \rightarrow \Delta PE = 0$$

$$V_1 = V_2 \rightarrow \Delta KE = 0$$

$$\Delta E = \Delta U$$

$$\Delta E = \Delta U + \cancel{\Delta KE} + \cancel{\Delta PE}$$

0 0

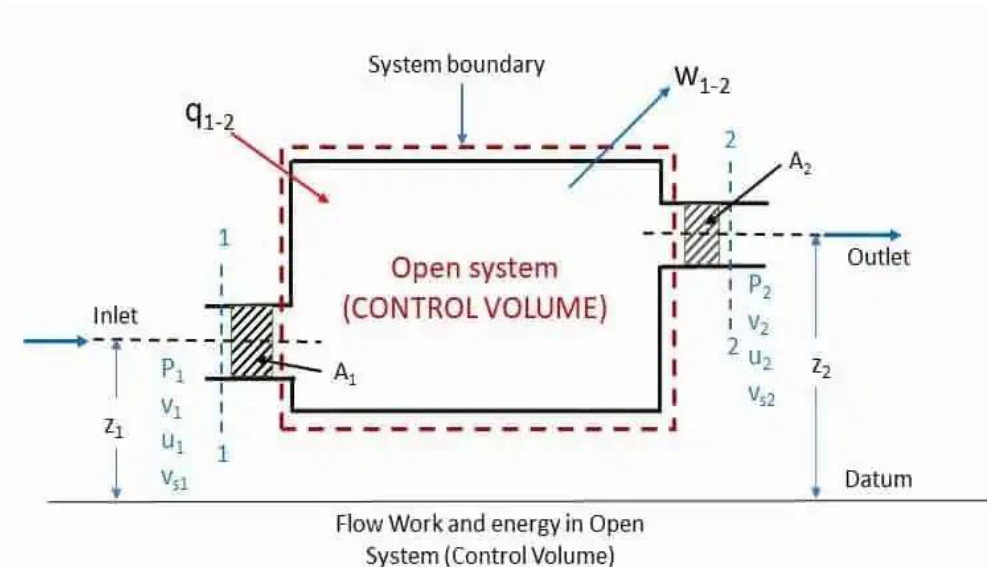


The energy change of a system during a process is equal to the net work and heat transfer between the system and its surroundings.

Mechanisms of Energy Transfer, E_{in} and E_{out}

- Energy can be transferred to or from a system in three forms: **heat**, **work**, and **mass flow**.
- Noting that energy can be transferred in the forms of heat, work, and mass, and that the net transfer of a quantity is equal to the difference between the amounts transferred in and out, the energy balance can be written more explicitly as

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta E_{system}$$



The First law of thermodynamics(Mathematical)

$$Q_{in} - Q_{out} + W_{in} - W_{out} = \Delta E$$

Assuming $Q_{in} > Q_{out}$ and $W_{out} > W_{in}$

$$Q_{net,in} - W_{net,out} = \Delta E$$

Usually we drop the subscripts, hence

$$Q - W = \Delta E \quad \text{General Form (KJ)}$$

$$\dot{Q} - \dot{W} = \Delta \dot{E} \quad \text{per unit time (or on a Rate basis) KJ/s= Watt}$$

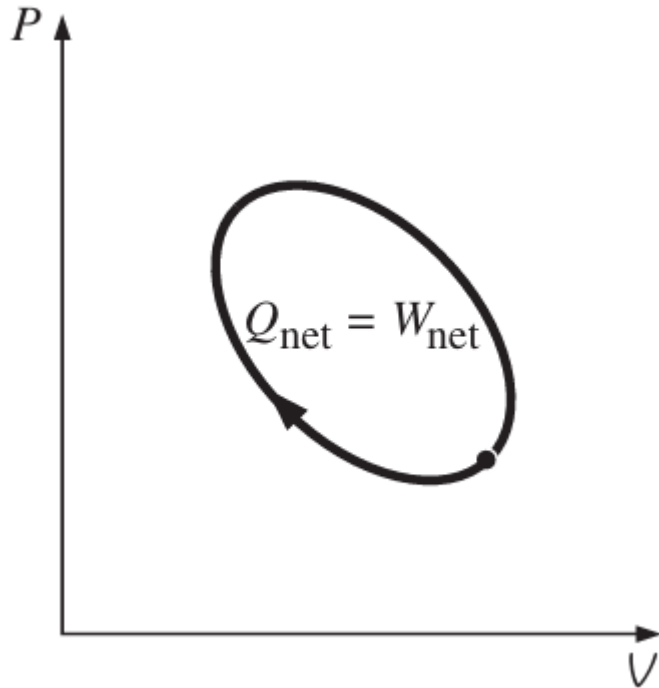
$$q - w = \Delta e \quad \text{per unit mass basis (KJ/kg)}$$

$$\delta q - \delta w = de \quad \text{differential form}$$

- Energy balance for any system undergoing **any kind of process** can be expressed more compactly as

$$\underbrace{E_{in} - E_{out}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc., energies}} \quad (\text{kJ})$$

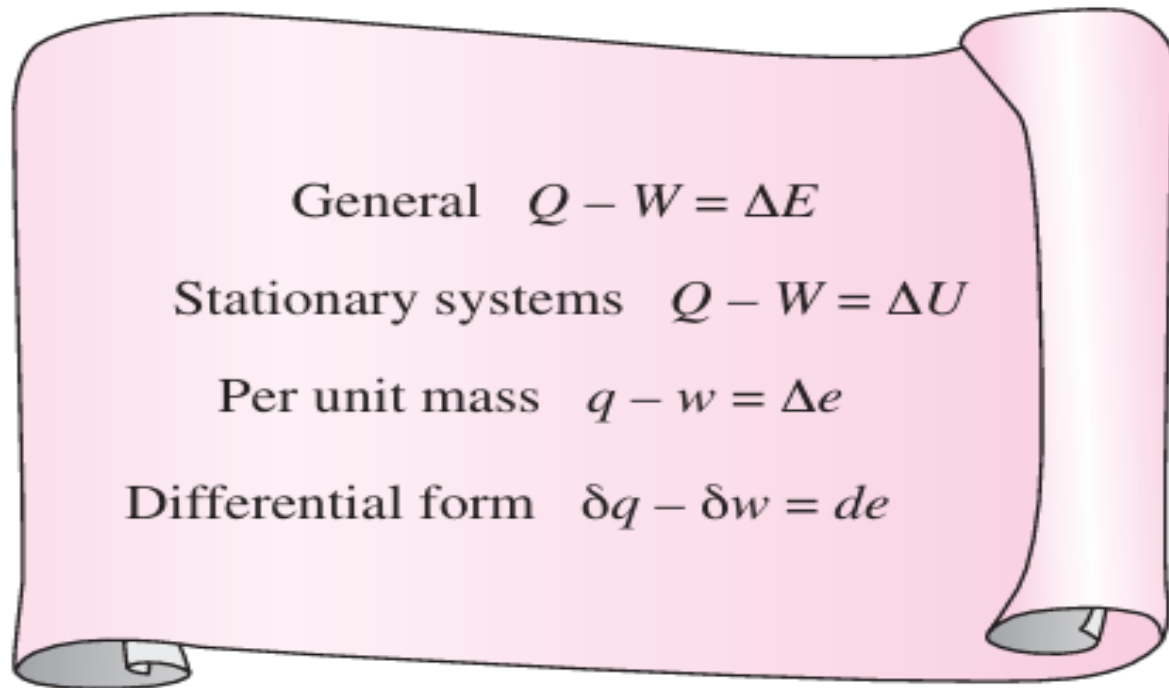
- For a closed system undergoing a cycle, the initial and final states are identical, and thus $\Delta E_{\text{system}} = E_2 - E_1 = 0$.



$$W_{\text{net,out}} = Q_{\text{net,in}} \quad \text{or} \quad \dot{W}_{\text{net,out}} = \dot{Q}_{\text{net,in}} \quad (\text{for a cycle})$$

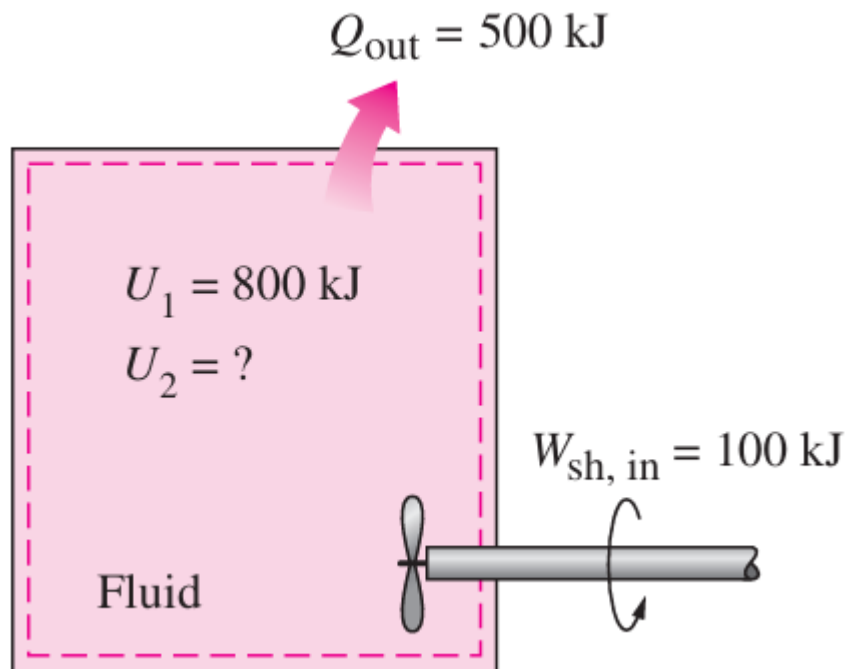
For a cycle $\Delta E = 0$, thus $Q = W$.

Various forms of the first-law relation for closed systems.



Example:4.1

A rigid tank contains a hot fluid that is cooled while being stirred by a paddle wheel. Initially, the internal energy of the fluid is 800 kJ. During the cooling process, the fluid loses 500 kJ of heat, and the paddle wheel does 100 kJ of work on the fluid. Determine the final internal energy of the fluid. Neglect the energy stored in the paddle wheel.



Solution:

Assumptions 1The tank is stationary and thus the kinetic and potential energy changes are zero, $\Delta KE = \Delta PE = 0$.

Therefore, $\Delta E = \Delta U$ and internal energy is the only form of the system's energy that may change during this process. 2Energy stored in the paddle wheel is negligible.

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc., energies}}$$

$$W_{\text{sh,in}} - Q_{\text{out}} = \Delta U = U_2 - U_1$$

$$100 \text{ kJ} - 500 \text{ kJ} = U_2 - 800 \text{ kJ}$$

$$U_2 = \mathbf{400 \text{ kJ}}$$

Therefore, the final internal energy of the system is 400 kJ.

Example:4.2

A well-insulated rigid tank contains 5 kg of a saturated liquid–vapor mixture of water at 100 kPa. Initially, three-quarters of the mass is in the liquid phase. An electric resistor placed in the tank is connected to a 110-V source, and a current of 8 A flows through the resistor when the switch is turned on. Determine how long it will take to vaporize all the liquid in the tank. Also, show the process on a T-v diagram with respect to saturation lines.

Solution:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$W_{e,\text{in}} = \Delta U = m(u_2 - u_1) \quad (\text{since } Q = \text{KE} = \text{PE} = 0)$$

$$VI\Delta t = m(u_2 - u_1)$$

The properties of water are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 100 \text{ kPa} \\ x_1 = 0.25 \end{array} \right\} \begin{array}{l} \nu_f = 0.001043, \quad \nu_g = 1.6941 \text{ m}^3/\text{kg} \\ u_f = 417.40, \quad u_{fg} = 2088.2 \text{ kJ/kg} \end{array}$$

$$\nu_1 = \nu_f + x_1 \nu_{fg} = 0.001043 + [0.25 \times (1.6941 - 0.001043)] = 0.42431 \text{ m}^3/\text{kg}$$

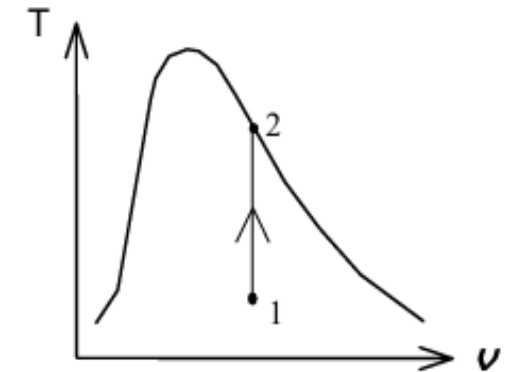
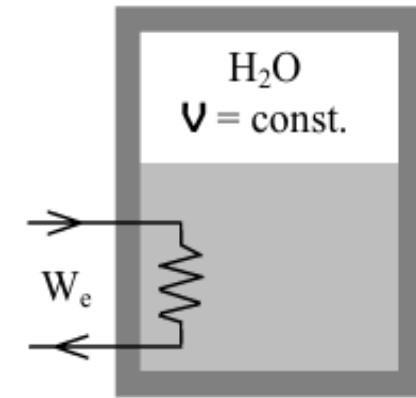
$$u_1 = u_f + x_1 u_{fg} = 417.40 + (0.25 \times 2088.2) = 939.4 \text{ kJ/kg}$$

$$\left. \begin{array}{l} \nu_2 = \nu_1 = 0.42431 \text{ m}^3/\text{kg} \\ \text{sat. vapor} \end{array} \right\} u_2 = u_{g@0.42431 \text{ m}^3/\text{kg}} = 2556.2 \text{ kJ/kg}$$

Substituting,

$$(110 \text{ V})(8 \text{ A})\Delta t = (5 \text{ kg})(2556.2 - 939.4) \text{ kJ/kg} \left(\frac{1000 \text{ VA}}{1 \text{ kJ/s}} \right)$$

$$\Delta t = 9186 \text{ s} \cong \mathbf{153.1 \text{ min}}$$



Example:4.3

An insulated piston–cylinder device contains 5 L of saturated liquid water at a constant pressure of 175 kPa. Water is stirred by a paddle wheel while a current of 8 A flows for 45 min through a resistor placed in the water. If one-half of the liquid is evaporated during this constant pressure process and the paddle-wheel work amounts to 400 kJ, determine the voltage of the source. Also, show the process on a P-v diagram with respect to saturation lines.

Solution:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$W_{e,\text{in}} + W_{\text{pw},\text{in}} - W_{b,\text{out}} = \Delta U \quad (\text{since } Q = \text{KE} = \text{PE} = 0)$$

$$W_{e,\text{in}} + W_{\text{pw},\text{in}} = m(h_2 - h_1)$$

$$(VI\Delta t) + W_{\text{pw},\text{in}} = m(h_2 - h_1)$$

since $\Delta U + W_b = \Delta H$ during a constant pressure quasi-equilibrium process. The properties of water are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 175 \text{ kPa} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} h_1 = h_{f@175 \text{ kPa}} = 487.01 \text{ kJ/kg} \\ v_1 = v_{f@175 \text{ kPa}} = 0.001057 \text{ m}^3/\text{kg} \end{array}$$

$$\left. \begin{array}{l} P_2 = 175 \text{ kPa} \\ x_2 = 0.5 \end{array} \right\} h_2 = h_f + x_2 h_{fg} = 487.01 + (0.5 \times 2213.1) = 1593.6 \text{ kJ/kg}$$

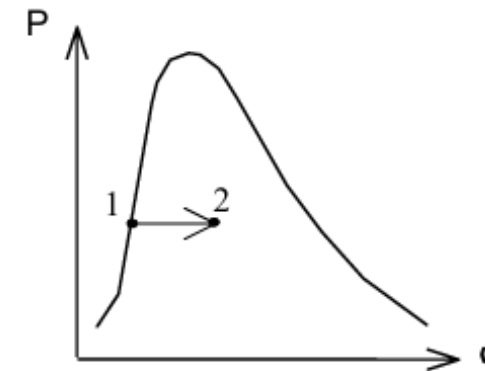
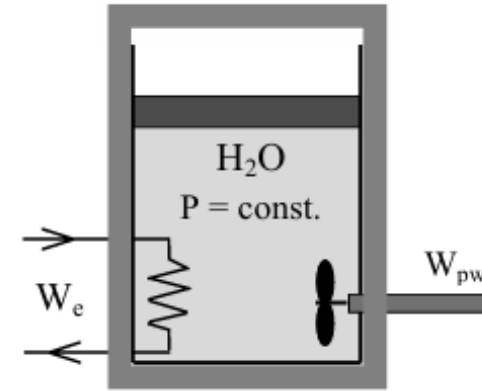
$$m = \frac{v_1}{v_1} = \frac{0.005 \text{ m}^3}{0.001057 \text{ m}^3/\text{kg}} = 4.731 \text{ kg}$$

Substituting,

$$VI\Delta t + (400 \text{ kJ}) = (4.731 \text{ kg})(1593.6 - 487.01) \text{ kJ/kg}$$

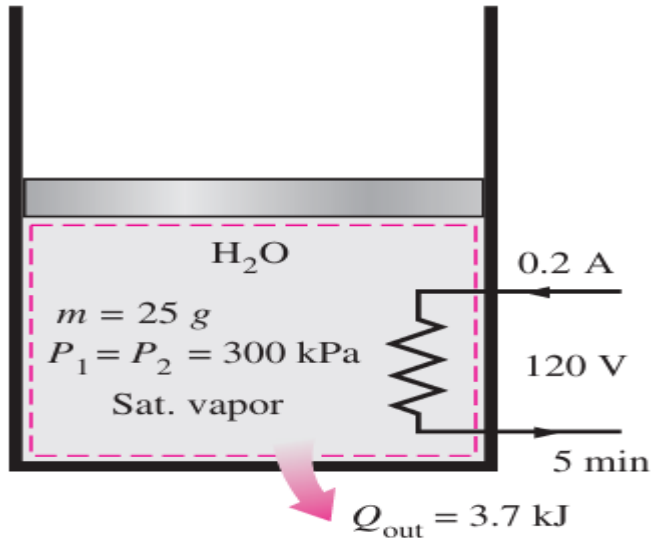
$$VI\Delta t = 4835 \text{ kJ}$$

$$V = \frac{4835 \text{ kJ}}{(8 \text{ A})(45 \times 60 \text{ s})} \left(\frac{1000 \text{ VA}}{1 \text{ kJ/s}} \right) = \mathbf{223.9 \text{ V}}$$



Example:4.4

A piston–cylinder device contains 25 g of saturated water vapor that is maintained at a constant pressure of 300 kPa. A resistance heater within the cylinder is turned on and passes a current of 0.2 A for 5 min from a 120-V source. At the same time, a heat loss of 3.7 kJ occurs. (a) Show that for a closed system the boundary work W_b and the change in internal energy ΔU in the first-law relation can be combined into one term, ΔH , for a constant pressure process. (b) Determine the final temperature of the steam.



Solution:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc., energies}}$$

$$Q - W = \Delta U + \cancel{\Delta KE}^0 + \cancel{\Delta PE}^0$$

$$Q - W_{\text{other}} - W_b = U_2 - U_1$$

For a constant-pressure process, the boundary work is given as $W_b = P_0(V_2 - V_1)$. Substituting this into the preceding relation gives

$$Q - W_{\text{other}} - P_0(V_2 - V_1) = U_2 - U_1$$

However,

$$P_0 = P_2 = P_1 \rightarrow Q - W_{\text{other}} = (U_2 + P_2 V_2) - (U_1 + P_1 V_1)$$

Also $H = U + PV$, and thus

$$Q - W_{\text{other}} = H_2 - H_1 \quad (\text{kJ})$$

(b) The only other form of work in this case is the electrical work, which can be determined from

$$W_e = \mathbf{VI} \Delta t = (120 \text{ V})(0.2 \text{ A})(300 \text{ s}) \left(\frac{1 \text{ kJ/s}}{1000 \text{ VA}} \right) = 7.2 \text{ kJ}$$

$$\text{State 1: } \left. \begin{array}{l} P_1 = 300 \text{ kPa} \\ \text{sat. vapor} \end{array} \right\} h_1 = h_g @ 300 \text{ kPa} = 2724.9 \text{ kJ/kg} \quad (\text{Table A-5})$$

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc., energies}}$$

$$W_{e,\text{in}} - Q_{\text{out}} - W_b = \Delta U$$

$$W_{e,\text{in}} - Q_{\text{out}} = \Delta H = m(h_2 - h_1) \quad (\text{since } P = \text{constant})$$

$$7.2 \text{ kJ} - 3.7 \text{ kJ} = (0.025 \text{ kg})(h_2 - 2724.9) \text{ kJ/kg}$$

$$h_2 = 2864.9 \text{ kJ/kg}$$

Now the final state is completely specified since we know both the pressure and the enthalpy. The temperature at this state is

$$\text{State 2:} \quad \left. \begin{array}{l} P_2 = 300 \text{ kPa} \\ h_2 = 2864.9 \text{ kJ/kg} \end{array} \right\} T_2 = \mathbf{200^\circ\text{C}} \quad (\text{Table A-6})$$

Example 4.5

Steam at 1100 kPa and 92 percent quality is heated in a rigid container until the pressure is 2000 kPa. For a mass of 0.05 kg, calculate the amount of heat supply (in kJ) and the total entropy change (in kJ/kg.K).

Solution:

State1

$$\text{at } P_1 = 1100 \text{ kPa}, x_1 = 0.92$$

$$\begin{aligned} v_1 &= v_{f1} + x_1 v_{fg1} \\ &= 0.00113 + 0.92(0.17753 - 0.001133) \\ &= 0.1634 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

$$\begin{aligned} u_1 &= u_{f1} + x_1 u_{fg1} \\ &= 779.78 + 0.92(1805.7) \\ &= 2441.024 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

$$\begin{aligned} s_1 &= s_{f1} + x_1 s_{fg1} \\ &= 2.1785 + 0.92(4.3735) \\ &= 6.20212 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

- ❖ For a rigid container,
 $v_2 = v_1 = 0.1634 \text{ m}^3/\text{kg}$

State2

at $P_2 = 2000 \text{ kPa}$, $v_2 = 0.1634 \frac{\text{m}^3}{\text{kg}}$

$v_2 > v_g$ superheated

v	u	s
0.15122	2945.9	7.1292
0.1634	u_2	s_2
0.17568	3116.9	7.4337

$$u_2 = 2945.9 + \left(\frac{0.1634 - 0.15122}{0.17568 - 0.15122} \right) (3116.9 - 2945.9)$$

$$= 3030.42 \frac{\text{kJ}}{\text{kg}}$$

$$s_2 = 7.1292 + \left(\frac{0.1634 - 0.15122}{0.17568 - 0.15122} \right) (7.4337 - 7.1292)$$

$$= 7.2790 \frac{\text{kJ}}{\text{kg.K}}$$

- ❖ Amount of heat supplied, Q

$$\begin{aligned}Q &= m(u_2 - u_1) \\&= 0.05(3030.42 - 2441.9) \\&= \underline{\underline{29.43 \text{ kJ}}}\end{aligned}$$

- ❖ The change in entropy, Δs

$$\begin{aligned}\Delta s &= s_2 - s_1 \\&= 7.2790 - 6.204 \\&= \underline{\underline{1.075 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}}}\end{aligned}$$



Example 4.6

A rigid tank is divided into two equal parts by a partition. Initially one side of the tank contains 5 kg water at 200 kPa and 25°C, and the other side is evacuated. The partition is then removed, and the water expands into the entire tank. The water is allowed to exchange heat with its surroundings until the temperature in the tank returns to the initial value of 25°C. Determine (a) the volume of the tank (b) the final pressure (c) the heat transfer for this process.

Solution:

State1

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa}, \\ T_1 = 25^\circ \text{C} \end{array} \right\} \begin{array}{l} T_1 < T_{sat} \text{ Comp. liquid} \\ v_1 = v_f @ 25^\circ \text{C} = 0.001003 \frac{\text{m}^3}{\text{kg}} \end{array}$$

initial volume of half reservoir

$$\begin{aligned} V_1 &= mv \\ &= 5(0.001003) \\ &= 0.005 \text{ m}^3 \end{aligned}$$

❖ The initial volume for entire tank

$$\begin{aligned} V_{\text{reservoir}} &= 2(0.005) \\ &= \underline{\underline{0.01 \text{ m}^3}} \end{aligned}$$

❖ The final pressure

State 2

$$\left. \begin{aligned} T_2 &= 25^\circ \text{C} \\ v_2 &= \frac{0.01}{5} = 0.002 \frac{\text{m}^3}{\text{kg}} \end{aligned} \right\} \begin{aligned} v_f &= 0.001003 \frac{\text{m}^3}{\text{kg}} \\ v_g &= 43.34 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

check region!

$v_f < v < v_g \rightarrow$ *saturated mixture*

then : $P_2 = P_{\text{sat}} = \underline{\underline{3.169 \text{ kPa}}}$

❖ The heat transfer for this process

$$Q_{\text{net}} - W_{\text{net}} = m(\Delta u + \Delta ke + \Delta Pe)$$

$$Q_{\text{net}} - \cancel{W_{\text{net}}} = m(\Delta u + \cancel{\Delta ke} + \cancel{\Delta Pe})$$

$$Q_{\text{net}} = m\Delta u = m(u_2 - u_1)$$

$$\rightarrow u_1 = u_{f @ 25^\circ \text{C}} = 104.83 \frac{\text{kJ}}{\text{kg}}$$

$$u_2 = u_f + x_2 u_{fg}$$

$$x_2 = \frac{v_2 - v_f}{v_{fg}}$$

$$= 2.3 \times 10^{-5}$$

$$\therefore u_2 = 104.83 + 2.3 \times 10^{-5} (2304.3)$$

$$= 104.88 \frac{\text{kJ}}{\text{kg}}$$

Then :

$$Q_{\text{net}} = 5 (104.88 - 104.83)$$

$$= \underline{\underline{0.25 \text{ kJ}}}$$

❖ +ve sign indicates heat transfer into the system.

Example 4.7

A rigid tank is divided into two equal parts by a partition. Initially one side of the tank contains 5 kg water at 200 kPa and 25°C, and the other side is evacuated. The partition is then removed, and the water expands into the entire tank. The water is allowed to exchange heat with its surroundings until the temperature in the tank returns to the initial value of 25°C. Determine (a) the volume of the tank (b) the final pressure (c) the heat transfer for this process.

Solution:

State1

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa}, \\ T_1 = 25^\circ \text{C} \end{array} \right\} \begin{array}{l} T_1 < T_{sat} \text{ Comp. liquid} \\ v_1 = v_f @ 25^\circ \text{C} = 0.001003 \frac{\text{m}^3}{\text{kg}} \end{array}$$

initial volume of half reservoir

$$\begin{aligned} V_1 &= mv \\ &= 5(0.001003) \\ &\square 0.005 \text{ m}^3 \end{aligned}$$

❖ The initial volume for entire tank

$$\begin{aligned} V_{\text{reservoir}} &= 2(0.005) \\ &= \underline{\underline{0.01 \text{ m}^3}} \end{aligned}$$

❖ The final pressure

State 2

$$\left. \begin{aligned} T_2 &= 25^\circ \text{C} \\ v_2 &= \frac{0.01}{5} = 0.002 \frac{\text{m}^3}{\text{kg}} \end{aligned} \right\} \begin{aligned} v_f &= 0.001003 \frac{\text{m}^3}{\text{kg}} \\ v_g &= 43.34 \frac{\text{m}^3}{\text{kg}} \end{aligned}$$

check region!

$v_f < v < v_g \rightarrow$ *saturated mixture*

then : $P_2 = P_{\text{sat}} = \underline{\underline{3.169 \text{ kPa}}}$

❖ The heat transfer for this process

$$Q_{\text{net}} - W_{\text{net}} = m(\Delta u + \Delta ke + \Delta Pe)$$

$$Q_{\text{net}} - \cancel{W_{\text{net}}} = m(\Delta u + \cancel{\Delta ke} + \cancel{\Delta Pe})$$

$$Q_{\text{net}} = m\Delta u = m(u_2 - u_1)$$

$$\rightarrow u_1 = u_{f @ 25^\circ \text{C}} = 104.83 \frac{\text{kJ}}{\text{kg}}$$

$$u_2 = u_f + x_2 u_{fg}$$

$$x_2 = \frac{v_2 - v_f}{v_{fg}}$$

$$= 2.3 \times 10^{-5}$$

$$\therefore u_2 = 104.83 + 2.3 \times 10^{-5} (2304.3)$$

$$= 104.88 \frac{\text{kJ}}{\text{kg}}$$

Then :

$$Q_{\text{net}} = 5 (104.88 - 104.83)$$

$$= \underline{\underline{0.25 \text{ kJ}}}$$

❖ +ve sign indicates heat transfer into the system.



Supplementary Problems 1

1. Two tanks are connected by a valve. Tank A contains 2 kg of carbon monoxide gas at 77°C and 0.7 bar. Tank B holds 8 kg of the same gas at 27°C and 1.2 bar. Then the valve is opened and the gases are allowed to mix while receiving energy via heat transfer from the surrounding. The final equilibrium temperature is found to be 42°C. Determine (a) the final pressure (b) the amount of heat transfer. Also state your assumption.
[$P_2 = 105 \text{ kPa}$, $Q = +37.25 \text{ kJ}$]
2. A piston cylinder device contains 0.2 kg of water initially at 800 kPa and 0.06 m³. Now 200 kJ of heat is transferred to the water while its pressure is held constant. Determine the final temperature of the water. Also, show the process on a T-V diagram with respect to saturation lines.
[721.1°C]

Supplementary Problems 1

3. A piston-cylinder device contains 6 kg of refrigerant-134a at 800 kPa and 50°C. The refrigerant is now cooled at constant pressure until it exist as a liquid at 24°C. Show the process on T-v diagram and determine the heat loss from the system. State any assumption made.
[1210.26 kJ]
4. A 0.5 m³ rigid tank contains refrigerant-134a initially at 200 kPa and 40 percent quality. Heat is now transferred to the refrigerant until the pressure reaches 800 kPa. Determine (a) the mass of the refrigerant in the tank and (b) the amount of heat transferred. Also, show the process on a P-v diagram with respect to saturation lines.
[12.3 kg, 2956.2 kJ]
5. An insulated tank is divided into two parts by a partition. One part of the tank contains 6 kg of an ideal gas at 50°C and 800 kPa while the other part is evacuated. The partition is now removed, and the gas expands to fill the entire tank. Determine the final temperature and the pressure in the tank.

[50°C, 400 kPa]



FIRST LAW OF THERMODYNAMICS

MASS & ENERGY ANALYSIS OF CONTROL VOLUME



Conservation of Mass

- Conservation of mass is one of the most fundamental principles in nature. We are all familiar with this principle, and it is not difficult to understand it!
- For closed system, the conservation of mass principle is implicitly used since the mass of the system remain constant during a process.
- However, for control volume, mass can cross the boundaries. So the amount of mass entering and leaving the control volume must be considered.



Mass and Volume Flow Rates

- Mass flow through a cross-sectional area per unit time is called the mass flow rate. Note the dot over the mass symbol indicates a time rate of change. It is expressed as

$$\dot{m} = \int \rho V \cdot dA$$

- If the fluid density and velocity are constant over the flow cross-sectional area, the mass flow rate is

$$\dot{m} = \rho AV = \frac{AV}{\nu}$$

$$\text{where } \nu = \frac{1}{\rho}$$

ν is called specific volume



Principal of Conservation of Mass

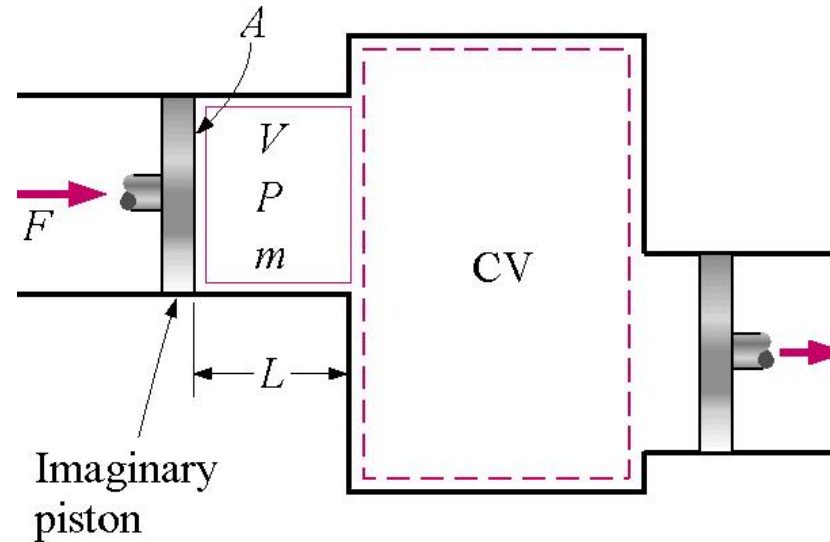
- The conservation of mass principle for a control volume can be expressed as

$$\dot{m}_{in} - \dot{m}_{out} = \dot{m}_{CV}$$

- For a steady state, steady flow process the conservation of mass principle becomes

$$\dot{m}_{in} = \dot{m}_{out} \quad (\text{kg/s})$$

Flow Work & The Energy of a Flowing Fluid

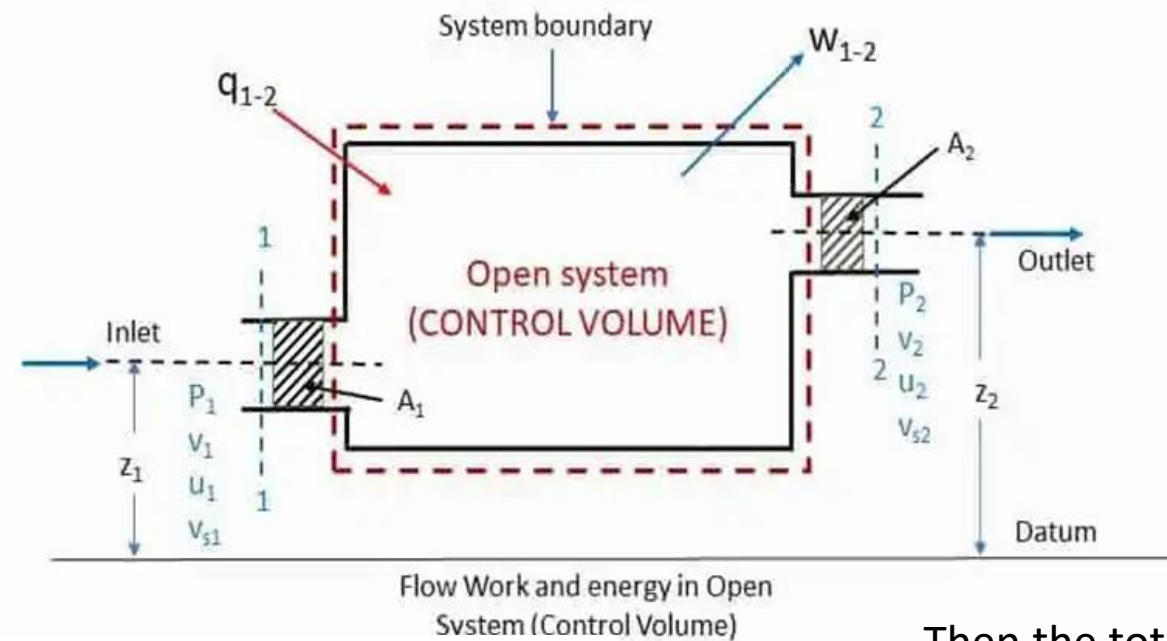


- As the fluid upstream pushes mass across the control volume, work done on that unit of mass is

$$\delta W_{flow} = F dL = F dL \frac{A}{A} = P dV = P v \delta m$$

$$\delta w_{flow} = \frac{\delta W_{flow}}{\delta m} = P v$$

This work is known as the flow work, or flow energy, and is necessary for maintaining a continuous flow through a control volume.



Flow energy

Kinetic energy

Internal energy

Potential energy

$$\theta = Pv + u + \frac{v^2}{2} + gz$$

Then the total energy of a flowing fluid on a unit-mass basis

$$\theta = Pv + e = Pv + (u + ke + pe)$$

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{system}}{dt}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \xrightarrow{0 \text{ (steady)}} = 0$$

First Law of Thermodynamics

$$\dot{Q}_{net} - \dot{W}_{net} + \sum \dot{m}_i \left(h_i + \frac{1}{2} v_i^2 + gz_i \right) - \sum \dot{m}_e \left(h_e + \frac{1}{2} v_e^2 + gz_e \right) = \frac{dE_{sys}}{dt} \xrightarrow{0 \text{ (steady)}}$$

Total Energy of a Flowing Fluid

- The total energy carried by a unit of mass as it crosses the control surface is the sum of the internal energy + flow work + potential energy + kinetic energy

$$\sum energy = u + P v + \frac{V^2}{2} + gz = h + \frac{V^2}{2} + gz$$

- The first law for a control volume can be written as

$$\dot{Q}_{net} - \dot{W}_{net} = \sum_{out} \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right) - \sum_{in} \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right)$$

Total Energy of a Flowing Fluid

- The steady state, steady flow conservation of mass and first law of thermodynamics can be expressed in the following forms
- For single-stream devices, the steady-flow energy balance equation becomes

$$q_{net} - w_{net} = \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2000} + \frac{g(z_2 - z_1)}{1000} \right) \quad (kJ / kg)$$

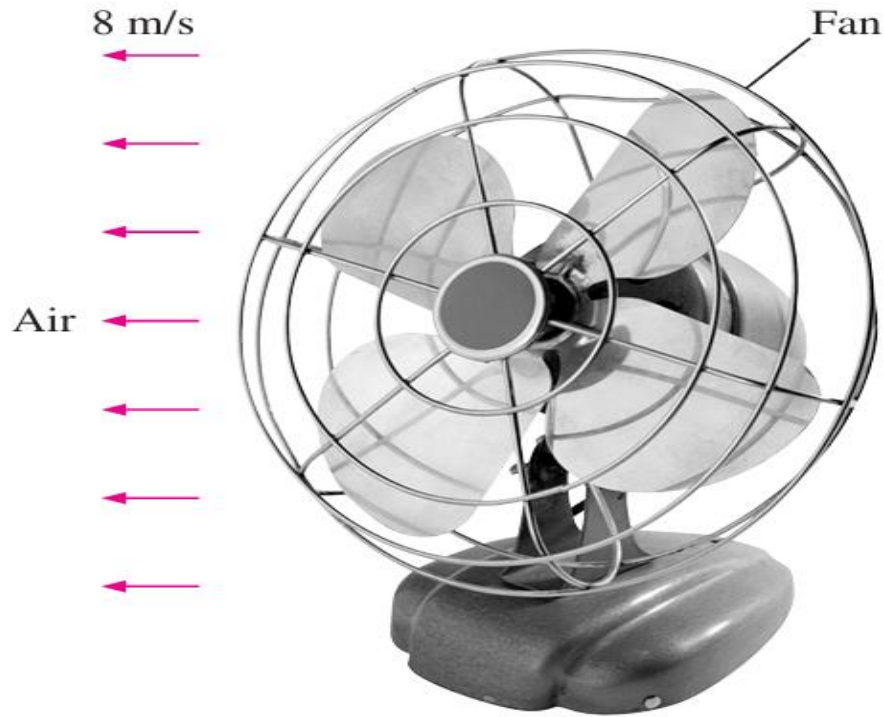
$$\left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right)$$

$$Q_{net} - W_{net} = m \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2000} + \frac{g(z_2 - z_1)}{1000} \right) \quad (kJ)$$

$$\dot{Q}_{net} - \dot{W}_{net} = \dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2000} + \frac{g(z_2 - z_1)}{1000} \right) \quad (kW)$$

Example:4.8

A fan that consumes 20 W of electric power when operating is claimed to discharge air from a ventilated room at a rate of 0.25 kg/s at a discharge velocity of 8 m/s . Determine if this claim is reasonable.



Solution:

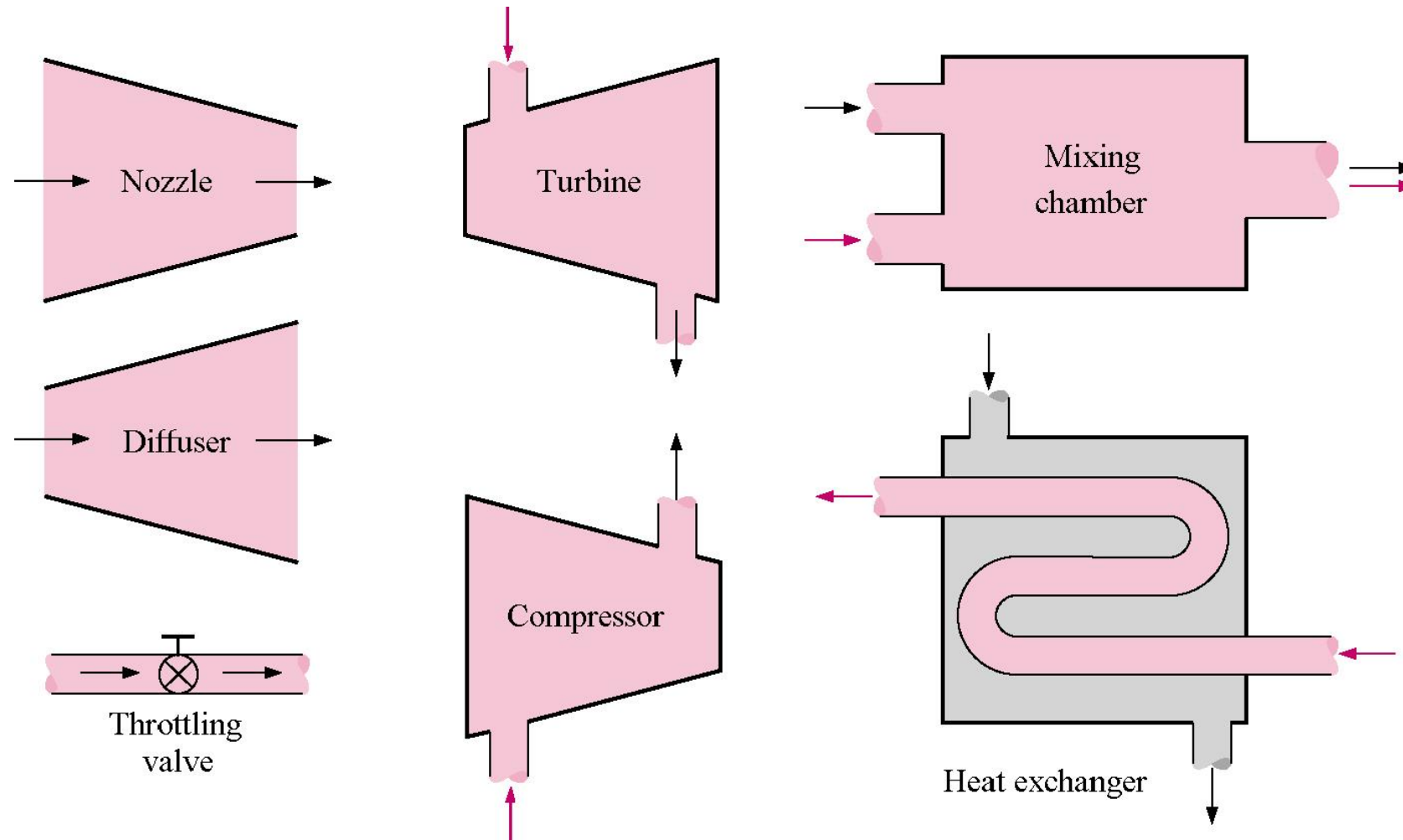
$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{\text{system}}}{dt} \rightarrow 0}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \text{ (steady)} = 0 \rightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$
$$\dot{W}_{\text{elect, in}} = \dot{m}_{\text{air}} \text{ke}_{\text{out}} = \dot{m}_{\text{air}} \frac{V_{\text{out}}^2}{2}$$

Solving for V_{out} and substituting gives the maximum air outlet velocity to be

$$V_{\text{out}} = \sqrt{\frac{\dot{W}_{\text{elect, in}}}{2\dot{m}_{\text{air}}}} = \sqrt{\frac{20 \text{ J/s}}{2(0.25 \text{ kg/s})} \left(\frac{1 \text{ m}^2/\text{s}^2}{1 \text{ J/kg}} \right)} = 6.3 \text{ m/s}$$

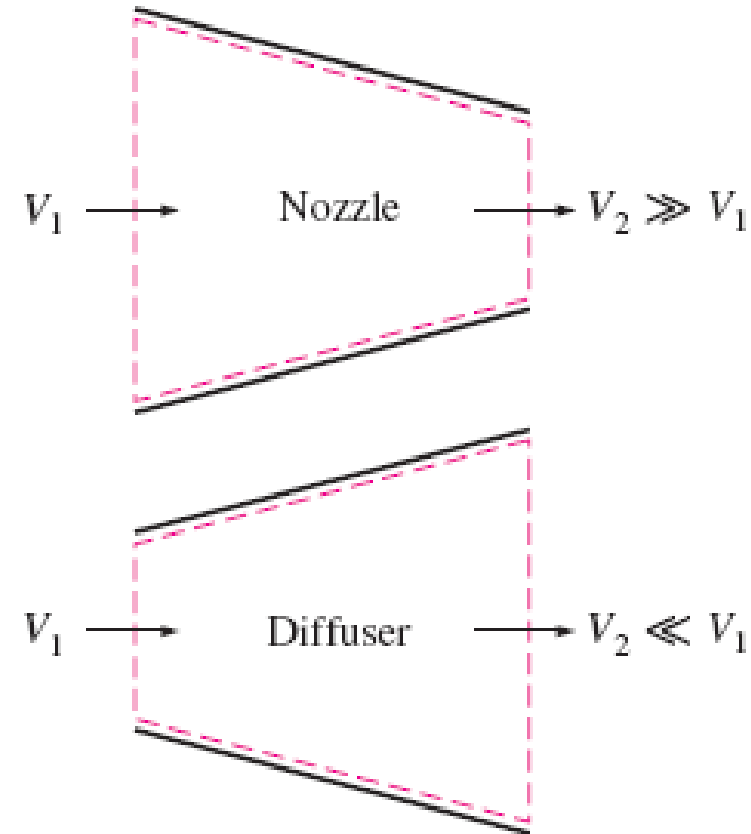
which is less than 8 m/s. Therefore, the claim is **false**.

Steady-flow Engineering Devices



Nozzle & Diffuser

- ❑ Nozzle - device that increases the velocity fluid at the expense of pressure.
- ❑ Diffuser - device that increases pressure of a fluid by slowing it down.
- ❑ Commonly utilized in jet engines, rockets, space-craft and even garden hoses.
- ❑ $Q = 0$ (heat transfer from the fluid to surroundings very small)
- ❑ $W = 0$ and $\Delta PE = 0$



□ Energy balance (nozzle & diffuser):

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in} \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right)$$

$$\dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} \right) = \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} \right)$$

$$\left(h_1 + \frac{V_1^2}{2} \right) = \left(h_2 + \frac{V_2^2}{2} \right)$$

Example 4.9

Steam at 0.4 MPa, 300°C, enters an adiabatic nozzle with a low velocity and leaves at 0.2 MPa with a quality of 90%. Find the exit velocity.

Solution:

State 1

$$P_1 = 0.4 \text{ MPa}$$

$$T_1 = 300^\circ \text{C}$$

$$V_1 \approx 0$$

State 2

$$P_2 = 0.2 \text{ MPa}$$

$$x_2 = 0.9$$

❖ Simplified energy balance:

$$\left(h_1 + \frac{V_1^2}{2} \right) = \left(h_2 + \frac{V_2^2}{2} \right)$$

State 1

$$\left. \begin{array}{l} P_1 = 0.4 \text{ MPa} \\ T_1 = 300^\circ \text{C} \end{array} \right\} \begin{array}{l} h_1 = 3067.1 \frac{\text{kJ}}{\text{kg}} \\ (\text{superheated}) \end{array}$$

State 2

$$\left. \begin{array}{l} P_2 = 0.2 \text{ MPa} \\ x_2 = 0.9 \end{array} \right\} \begin{array}{l} h_2 = h_f + x_2 h_{fg} \\ h_2 = 2486.1 \frac{\text{kJ}}{\text{kg}} \end{array}$$

❖ Exit velocity:

$$\begin{aligned} V_2 &= \sqrt{2000(3067.1 - 2486.1)} \\ &= \underline{\underline{1078 \text{ m/s}}} \end{aligned}$$

Example 4.10

Air at 10°C and 80 kPa enters the *diffuser* of a jet engine steadily with a velocity of 200 m/s . The inlet area of the diffuser is 0.4 m^2 . The air leaves the diffuser with a velocity that is very small compared with the inlet velocity. Determine (a) the mass flow rate of the air and (b) the temperature of the air leaving the diffuser.

State 1

$$P_1 = 80\text{ kPa}$$

$$T_1 = 10^{\circ}\text{C}$$

$$V_1 = 200\text{ m/s}$$

$$A_1 = 0.4\text{ m}^2$$

State 2

$$V_2 \approx 0$$

Solution:

❖ Simplified energy balance:

$$\left(h_1 + \frac{V_1^2}{2} \right) = \left(h_2 + \frac{V_2^2}{2} \right)$$

❖ From Ideal Gas Law:

$$v_1 = \frac{RT_1}{P_1} = 1.015 \frac{\text{m}^3}{\text{kg}}$$

❖ Mass flow rate

$$\begin{aligned}
 \dot{m} &= \frac{1}{v_1} V_1 A_1 \\
 &= \left(\frac{1}{1.015} \right) (200)(0.4) \\
 &= \underline{\underline{78.8 \frac{\text{kg}}{\text{s}}}}
 \end{aligned}$$

❖ Enthalpy at state 1

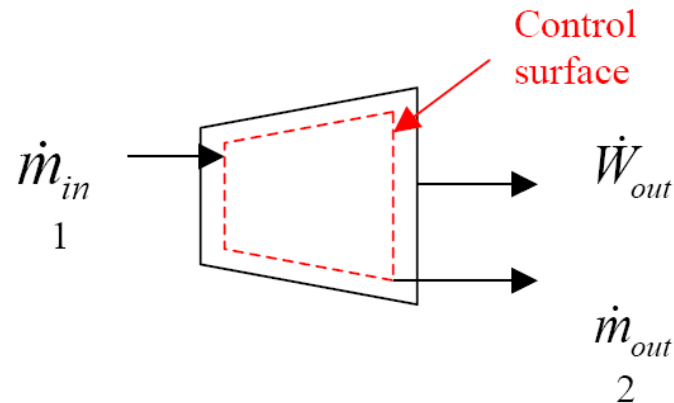
$$\begin{aligned}
 h_1 &= C_p T_1 = 1.005(283) \\
 &= 284.42 \frac{\text{kJ}}{\text{kg}}
 \end{aligned}$$

❖ From energy balance:

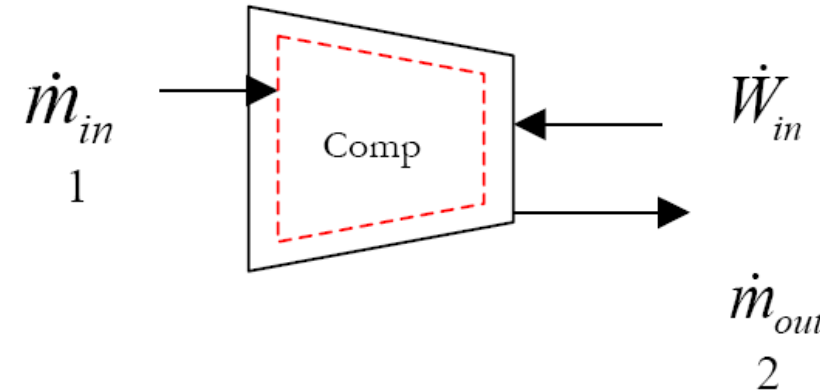
$$\begin{aligned}
 h_2 &= h_1 + \frac{V_1^2}{2000} \\
 &= 284.42 + \frac{200^2}{2000} \\
 &= 304.42 \frac{\text{kJ}}{\text{kg}}
 \end{aligned}$$

$$\begin{aligned}
 T_2 &= \frac{h_2}{C_p} \\
 &= \frac{304.42}{1.005} \\
 &= \underline{\underline{302.9 \text{ K}}}
 \end{aligned}$$

Turbine & Compressor



Turbine control volume



Steady-Flow Compressor

- ❑ Turbine – a work producing device through the expansion of a fluid.
- ❑ Compressor (as well as pump and fan) - device used to increase pressure of a fluid and involves work input.
- ❑ $Q = 0$ (well insulated), $\Delta PE = 0$, $\Delta KE = 0$ (very small compare to Δ enthalpy).

□ Energy balance: for turbine

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in} \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right)$$

$$\dot{m}_{in} (h_{in}) = \dot{W}_{out} + \dot{m}_{out} (h_{out})$$

$$\dot{W}_{out} = \dot{m} (h_1 - h_2)$$

□ Energy balance: for compressor, pump and fan

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in} \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right)$$

$$\dot{W}_{in} + \dot{m}_{in} (h_{in}) = \dot{m}_{out} (h_{out})$$

$$\dot{W}_{in} = \dot{m} (h_2 - h_1)$$



Example 4.11

The power output of an adiabatic steam turbine is 5 MW. Compare the magnitudes of Δh , Δke , and Δpe . Then determine the work done per unit mass of the steam flowing through the turbine and calculate the mass flow rate of the steam.

Data	:	Inlet ($P = 2 \text{ MPa}$, $T = 400^\circ\text{C}$, $v = 50 \text{ m/s}$, $z = 10 \text{ m}$)
		Exit ($P = 15 \text{ kPa}$, $x = 90\%$, $v = 180 \text{ m/s}$, $z = 6 \text{ m}$)

Solution:

State 1

$$\left. \begin{array}{l} p_1 = 2 \text{ MPa} \\ T_1 = 400^\circ \text{C} \end{array} \right\} \text{superheated} \quad h_1 = 3248.4 \frac{\text{kJ}}{\text{kg}}$$

State 2

$$\left. \begin{array}{l} P_2 = 15 \text{ kPa} \\ x_2 = 0.9 \end{array} \right\} \text{sat. mixture}$$

$$\begin{aligned} h_2 &= h_{f2} + x_2 h_{fg2} \\ &= 225.94 + 0.9 (2372.3) \\ &= 2361.01 \frac{\text{kJ}}{\text{kg}} \end{aligned}$$

❖ From energy balance:

$$\cancel{\dot{Q}}_{in} + \cancel{\dot{W}}_{in} + \sum_{in} \dot{m}_{in} \left(h_{in} + \frac{V_{in}^2}{2} + gz_{in} \right) = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \dot{m}_{out} \left(h_{out} + \frac{V_{out}^2}{2} + gz_{out} \right)$$

❖ Solve the equation:

$$\Delta h = h_2 - h_1 = -887.39 \frac{\text{kJ}}{\text{kg}}$$

$$\Delta KE = \frac{V_2^2 - V_1^2}{2000} = 14.95 \frac{\text{kJ}}{\text{kg}}$$

$$\Delta PE = \frac{g(z_2 - z_1)}{1000} = -0.04 \frac{\text{kJ}}{\text{kg}}$$

❖ the work done per unit mass

$$\begin{aligned} W_{out} &= \left[(h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2000} \right) + \left(\frac{g(z_1 - z_2)}{1000} \right) \right] \\ &= 887.39 - 14.95 + 0.04 \\ &= \underline{\underline{872.48 \frac{kJ}{kg}}} \end{aligned}$$

❖ The mass flow rate

$$\dot{m} = \frac{\dot{W}_{out}}{W_{out}} = \frac{5000}{872.48} = \underline{\underline{5.73 \frac{kg}{s}}}$$

Example 4.12

Air at 100 kPa and 280 K is compressed steadily to 600 kPa and 400 K. The mass flow rate of the air is 0.02 kg/s, and a heat loss of 16 kJ/kg occurs during the process. Assuming the changes in kinetic and potential energies are negligible, determine the necessary power input to the compressor.

Solution:

❖ simplified energy balance:

$$\begin{aligned}\dot{W}_{in} &= \dot{m}(h_2 - h_1) + \dot{Q}_{out} \\ &= \dot{m}(h_2 - h_1) + \dot{m}q_{out}\end{aligned}$$

State 1

$$\left. \begin{array}{l} P_1 = 100 \text{ kPa} \\ T_1 = 280 \text{ K} \end{array} \right\} \begin{array}{l} \text{air} \\ h_1 = 280.13 \frac{\text{kJ}}{\text{kg}} \end{array}$$

$h_1 = h @ T_1$
Table A-17

State 2

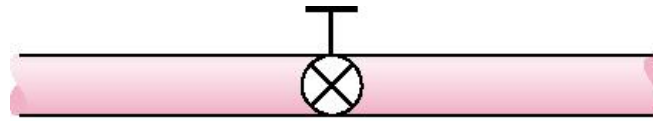
$$\left. \begin{array}{l} P_2 = 600 \text{ kPa} \\ T_2 = 400 \text{ K} \end{array} \right\} \begin{array}{l} \text{air} \\ h_2 = 400.98 \frac{\text{kJ}}{\text{kg}} \end{array}$$

$h_2 = h @ T_2$
Table A-17

❖ Thus

$$\begin{aligned}\dot{W}_{in} &= 0.02 \left[(400.98 - 280.13) + 16 \right] \\ &= \underline{\underline{2.74 \text{ kW}}}\end{aligned}$$

Throttling Valve



(a) An adjustable valve



(b) A porous plug



(c) A capillary tube

Flow-restricting devices that cause a significant pressure drop in the fluid.

Some familiar examples are ordinary adjustable valves and capillary tubes.

Example 4.13

State 1

$$\left. \begin{array}{l} P_1 = 8000 \text{ kPa} \\ T_1 = 300^\circ \text{C} \end{array} \right\} \begin{array}{l} \text{superheated} \\ h_1 = 2786.5 \frac{\text{kJ}}{\text{kg}} \end{array}$$

State 2

$$\left. \begin{array}{l} P_2 = 1600 \text{ kPa} \\ h_2 = h_1 \end{array} \right\} \text{make interpolation}$$

Steam enters a throttling valve at 8000 kPa and 300°C and leaves at a pressure of 1600 kPa. Determine the final temperature and specific volume of the steam.

$P(\text{kPa})$	$T(^{\circ}\text{C})$	v_f	v_g	h_f	h_g
1500	198.29	0.001154	0.131710	844.55	2791
1600	T_2	v_{f2}	v_{g2}	h_{f2}	h_{g2}
1750	205.72	0.001166	0.113440	878.16	2795.2



- At state 2, the region is sat. mixture
- Specific volume at state 2

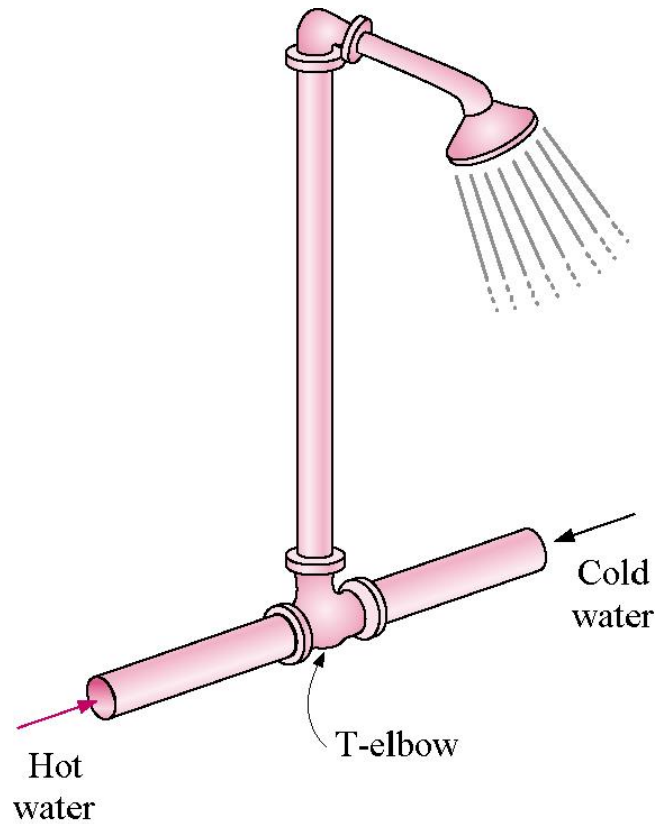
$$T_2 = T_{sat} = \underline{\underline{201.3^\circ C}}$$

- Getting the quality at state 2

$$\begin{aligned} x_2 &= \frac{h_2 - h_{f2}}{h_{g2} - h_{f2}} \\ &= \frac{2786.5 - 857.994}{2792.68 - 857.994} \\ &= 0.997 \end{aligned}$$

$$\begin{aligned} v_2 &= v_{f2} + x_2 v_{fg2} \\ &= 0.0011588 + \\ &\quad 0.997(0.124402 - 0.0011588) \\ &= \underline{\underline{0.1240 \frac{m^3}{kg}}} \end{aligned}$$

Mixing Chamber



The section where the mixing process takes place.

An ordinary T-elbow or a Y-elbow in a shower, for example, serves as the mixing chamber for the cold- and hot-water streams.

Mixing Chamber

□ Energy Balance:

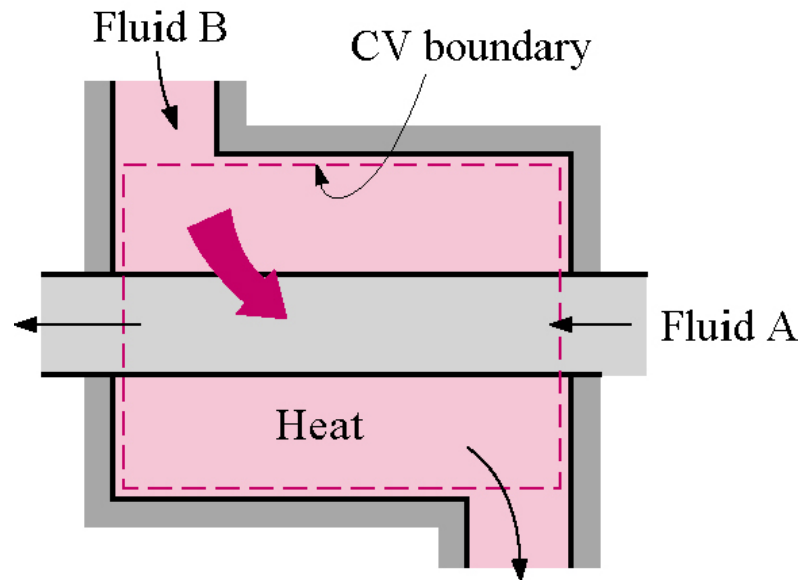
$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3$$

$$\dot{m}_1 h_1 + (\dot{m}_3 - \dot{m}_1) h_2 = \dot{m}_3 h_3$$

$$\dot{m}_1 (h_1 - h_2) = \dot{m}_3 (h_3 - h_2)$$

$$\dot{m}_1 = \dot{m}_3 \left(\frac{h_3 - h_2}{h_1 - h_2} \right)$$

Heat Exchanger



(a) System: Entire heat exchanger ($Q_{CV} = 0$)

Devices where two moving fluid streams exchange heat without mixing.

Heat exchangers typically involve no work interactions ($w = 0$) and negligible kinetic and potential energy changes for each fluid stream.

Example 4.14

Liquid sodium, flowing at 100 kg/s, enters a heat exchanger at 450°C and exits at 350°C. The specific heat of sodium is 1.25 kJ/kg.°C. Water enters at 5000 kPa and 20°C. Determine the minimum mass flux of the water so that the water does not completely vaporize. Neglect the pressure drop through the exchanger. Also, calculate the rate of heat transfer.

Solution:

❖ simplified energy balance:

$$\dot{m}_s h_{1s} + \dot{m}_w h_{1w} = \dot{m}_s h_{2s} + \dot{m}_w h_{2w}$$

$$\dot{m}_s (h_{1s} - h_{2s}) = \dot{m}_w (h_{2w} - h_{1w})$$

$$\dot{m}_s C_{p,s} (T_{1s} - T_{2s}) = \dot{m}_w (h_{2w} - h_{1w})$$

State 1: water

$$\left. \begin{array}{l} P_1 = 5000 \text{ kPa} \\ T_1 = 20^\circ \text{C} \end{array} \right\} \text{comp. liquid} \quad h_{1w} = 88.61 \frac{\text{kJ}}{\text{kg}}$$

State 2: water

$$P_2 = 5000 \text{ kPa}$$

$$h_{2w} = 2794.2 \frac{\text{kJ}}{\text{kg}}$$

Assume a sat. vapor state to obtain the max. allowable exiting enthalpy.

- ❑ the minimum mass flux of the water so that the water does not completely vaporize

$$\begin{aligned}\dot{m}_w &= \frac{m_s C_{p,s} (T_{1s} - T_{2s})}{(h_{2w} - h_{1w})} \\ &= \frac{100(1.25)(450 - 350)}{2794.2 - 88.61} \\ &= \underline{\underline{4.62 \frac{kg}{s}}}\end{aligned}$$

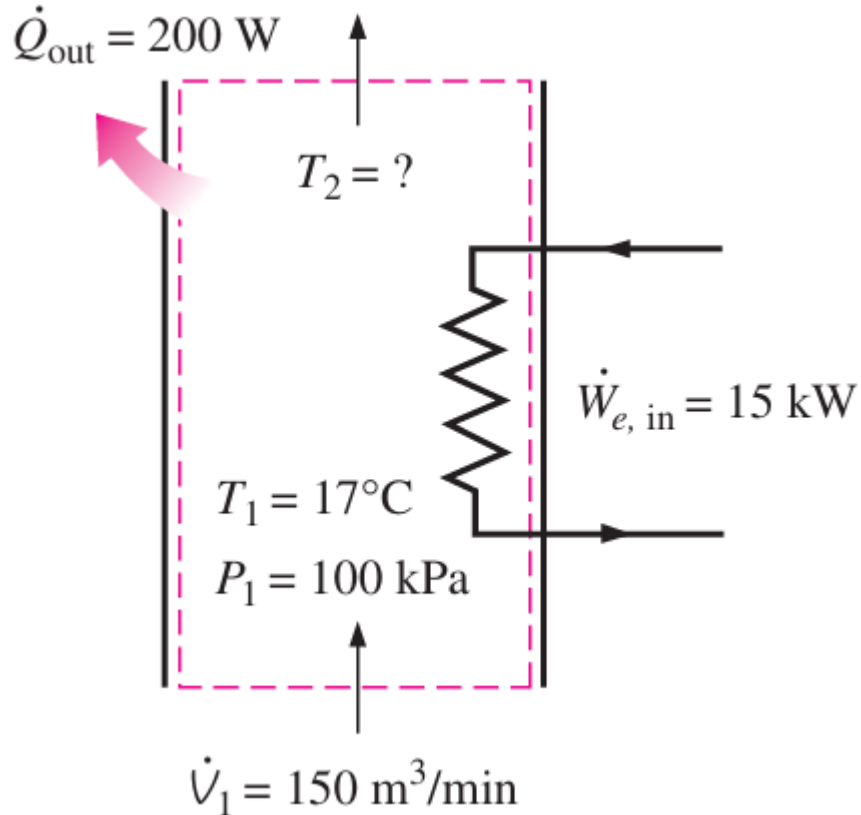
- ❑ the rate of heat transfer

$$\begin{aligned}\dot{Q}_w &= \dot{m}_w (h_{2w} - h_{1w}) \\ &= 4.62(2794.2 - 88.61) \\ &= \underline{\underline{12.5 MW}}\end{aligned}$$

Example: 4.15

The electric heating systems used in many houses consist of a simple duct with resistance heaters. Air is heated as it flows over resistance wires. Consider a 15-kW electric heating system. Air enters the heating section at 100 kPa and 17°C with a volume flow rate of 150 m³/min. If heat is lost from the air in the duct to the surroundings at a rate of 200 W, determine the exit temperature of air.

Solution:



$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{\text{system}}/dt}_{\text{Rate of change in internal, kinetic, potential, etc., energies}}} \overset{0 \text{ (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{e, \text{in}} + \dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{W}_{e, \text{in}} - \dot{Q}_{\text{out}} = \dot{m}c_p(T_2 - T_1)$$

From the ideal-gas relation, the specific volume of air at the inlet of the duct is

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(290 \text{ K})}{100 \text{ kPa}} = 0.832 \text{ m}^3/\text{kg}$$

The mass flow rate of the air through the duct is determined from

$$\dot{m} = \frac{\dot{V}_1}{v_1} = \frac{150 \text{ m}^3/\text{min}}{0.832 \text{ m}^3/\text{kg}} \left(\frac{1 \text{ min}}{60 \text{ s}} \right) = 3.0 \text{ kg/s}$$

Substituting the known quantities, the exit temperature of the air is determined to be

$$(15 \text{ kJ/s}) - (0.2 \text{ kJ/s}) = (3 \text{ kg/s})(1.005 \text{ kJ/kg} \cdot ^\circ\text{C})(T_2 - 17)^\circ\text{C}$$

$$T_2 = \mathbf{21.9^\circ\text{C}}$$

Discussion Note that heat loss from the duct reduces the exit temperature of air.



Supplementary Problems 2

1. Air flows through the supersonic nozzle . The inlet conditions are 7 kPa and 420°C. The nozzle exit diameter is adjusted such that the exiting velocity is 700 m/s. Calculate (a) the exit temperature, (b)the mass flux, and (c) the exit diameter. Assume an adiabatic quasiequilibrium flow.
2. Steam at 5 MPa and 400°C enters a nozzle steadily velocity of 80 m/s, and it leaves at 2 MPa and 300°C. The inlet area of the nozzle is 50 cm², and heat is being lost at a rate of 120 kJ/s. Determine (a) the mass flow rate of the steam, (b) the exit velocity of the steam, and (c) the exit area nozzle.
3. Steam enters a turbine at 4000 kPa and 500°C and leaves as shown in Fig A below. For an inlet velocity of 200 m/s, calculate the turbine power output. (a)Neglect any heat transfer and kinetic energy change (b)Show that the kinetic energy change is negligible.

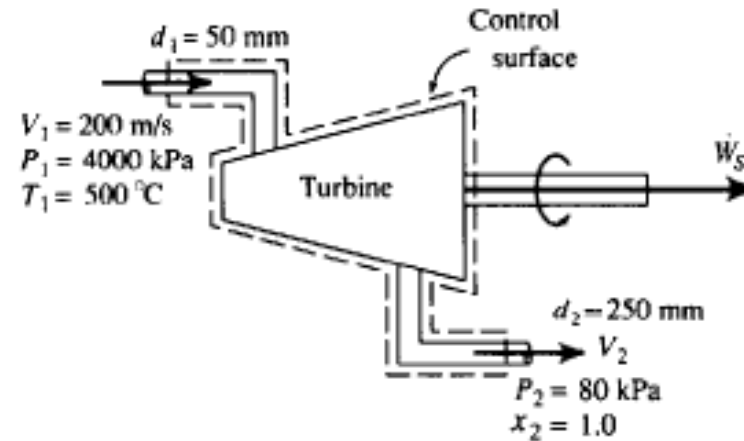


Figure A

4. Consider an ordinary shower where hot water at 60°C is mixed with cold water at 10°C . If it is desired that a steady stream of warm water at 45°C be supplied, determine the ratio of the mass flow rates of the hot to cold water. Assume the heat losses from the mixing chamber to be negligible and the mixing to take place at a pressure of 150 kPa .
5. Refrigerant-134a is to be cooled by water in a condenser. The refrigerant enters the condenser with a mass flow rate of 6 kg/min at 1 MPa and 70°C and leaves at 35°C . The cooling water enters at 300 kPa and 15°C and leaves at 25°C . Neglecting any pressure drops, determine (a) the mass flow rate of the cooling water required and (b) the heat transfer rate from the refrigerant to water.

THANK YOU